MIL-C-24643/8B
3 October 1986
SUPERSEDING
MIL-C-0024643/8A(SH)
8 November 1985
MIL-C-24643/8
28 September 1984

MILITARY SPECIFICATION SHEET

CORD, ELECTRICAL, 300 VOLTS, TYPE LSMMOP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cord described herein shall consist of this specification and the latest issue of MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, AWG size 24, class 0 stranding, uncoated.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Ethylene propylene rubber or cross-linked polyethylene insulation, nominal wall thickness 0.020 inch. Standard identification code applied by method 3.
Fourth - Five conductors cabled together with a lay not greater than 2 inches. A tie cord shall be laid straight as the central core. Fillers shall be used as necessary to make a firm, round cord. Cabling sequence shall be consecutive.
Fifth - A binder of polyester tape or braid.
Sixth - Cross-linked polyolefin jacket.

TABLE I. Details.

Military part no. M24643/8	Type and size	Number of conductors	Insulated conductor minimum (inch)	Overall maximum (inch)
-01UN	LSMMOP-5	5	0.065	0.305

(B) denotes changes.

AMSC N/A FSC 6145
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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet at 25°C, maximum	27.96
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	1000
Insulation resistance - megohms/1000 feet, minimum	100
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Group B:	
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Ethylene propylene rubber	
Tensile strength - lb/in^2, minimum	700
Elongation - percent, minimum	250
Cross-linked polyethylene	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cord)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Twisting and bending endurance (bending endurance mandrel 0.8-inch diameter, maximum)	
At 75 ± 2°C, cycles, minimum	6000
At minus 30 ± 2°C, cycles, minimum	2000
Cold bending (cord) - at minus 40 ± 2°C	No failure
Group C:	
Physicals (aged)	
Insulation	
Ethylene propylene rubber	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged, minimum	75
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80

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EXAMINATION AND TESTS: (Continued)

	Requirements
Jacket (cord)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (jacket) - cycles, minimum	250
Permanence of printing (conductor) - cycles, minimum	50
Cable filler removability	No failure
Heat distortion - percent of unaged, maximum	30

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, and C, plus the following:

	Aging and compatibility (cord)(125 ± 5°C)	No failure
(B)	Abrasion resistance (jacket) - scrapes, minimum	75
	Acid gas equivalent - percent, maximum	
	Jacket	2
	Fillers	2
	Insulation	18
	Halogen content - percent, maximum	
	Jacket	0.2
	Fillers	0.2
	Flame propagation (cord)	No failure
	Immersion (jacket)	
	Tensile strength - percent of unaged, minimum	50
	Elongation - percent of unaged, minimum	50
	Smoke index, maximum	
	Jacket	25
	Fillers	—
(B)	Insulation	45
	Toxicity index, maximum	
	Jacket	5
	Fillers	5
	Insulation	1.5
	Durometer (jacket) - (type A) hardness, minimum	80
	Weathering (jacket)	No failure

UNIT ORDERING LENGTH: 500 feet (nominal)

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Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-08)

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